



Latest Trends In The Consumer Electronics Testing Scenario

Automation of electronics testing will help manufacturers save significantly in terms of costs and manpower. This will bring about a digital transformation in the electronics appliance testing segment



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With reduction in size of electronic gadgets having wireless capabilities, testing has become a vital process in the product development cycle. Testing and measuring instruments are required for timely elimination of defects and prevent equipment failure in consumer electronics. Testing is crucial at every stage of the production process, whether manufacturing, maintenance, or research and development.

There are several emerging consumer electronics divisions that require innovative and latest testing solutions to meet the demands of integration of new technologies into them. A few such sectors are automotive, data centres, power applications, the Internet of Things (IoT) devices, electrical appliances and 5G.

A high number of components in a small package creates issues like signal interference, inductive or capacitive coupling, improper functioning of individual components, etc. Therefore every electronic product needs to be tested multiple times, using diverse test equipment for each component. The challenge is to identify the type of test equipment needed for the desired testing.

Testing and certification of electronic devices is done at different levels, namely, system, field, inter-

Consumer electronics testing (Credit: www.csagroup.org)